Integrating Multi-Source Spatial Data into a Graph Framework for Carbon Storage Suitability Analysis

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Table A1. Layers and Data Sources Descriptions.

Layer Name	Database Description	Node Type	Database Source
LA_CO2_Source	CO2 Emission Facilities Points	Source Nodes	U.S. Environmental Protection Agency. (2010). Facility Level Information on Greenhouse Gases Tool (FLIGHT). Retrieved from https://ghgdata.epa.gov/ghgp/main.do
LA_CO2Pipe	CO2 Denbury Pipeline Polyline	Source Edges	Denbury Resources. Denbury CO_2 Pipeline. Esri. Retrieved from https://www.arcgis.com/home/item.html?id=692dddd98c8742488c14828d0663da58
LA_EPA_EJScreen	EPA Environmental Justice Screening 2024 Shapefile	Socioeconomic Nodes	United States Environmental Protection Agency. Environmental Justice Screening Tool (EJScreen). 2024, from https://www.epa.gov/ejscreen/download-ejscreen-data. **Note:** The EPA's EJScreen platform is no longer publicly accessible. However, archived collections of the underlying variables remain available for select regions and for replication purposes at: https://pedp-ejscreen.azurewebsites.net/
LA_LANDGRID_Parcels	LANDGRID Parcel Dataset Shapefile	Socioeconomic Nodes	US Land Grid Parcel Data
LA_COAST_Zone	Louisiana Coastal Zone Boundary Shapefile	Socioeconomic Nodes	Louisiana Department of Energy and Natural Resources, Office of Coastal Management. (2012). Louisiana Coastal Zone Boundary. https://www.dnr.louisiana.gov/page/coastal-zone-boundary-2012
LA_Tribal_land	Federally Recognized Tribal Land Polygon	Socioeconomic Nodes	U.S. Census Bureau. (2020). TIGER/Line Shapefile, 2020, Nation, U.S., American Indian Tribal Subdivisions [Data set]. U.S. Department of Commerce. https://catalog.data.gov/dataset/tiger-line-shapefile-2020-nation-u-s-american-indian-tribal-subdivisions
LA_Pet_Reserve	Strategic Petroleum Reserves Polygons	Socioeconomic Nodes	U.S. Energy Information Administration. (2022). Strategic Petroleum Reserves [Data set]. U.S. Department of Energy. Retrieved from https://atlas.eia.gov/datasets/strategic-petroleum-reserves-1

LA_Pop_Cluster	Population 2020 Census tracts shapefile	Socioeconomic Nodes	U.S. Census Bureau. (2021). TIGER/Line Shapefile, 2020, Nation, U.S., Census Tracts. U.S. Department of Commerce. https://www.census.gov/geographies/mapping-files/2020/geo/tiger-line-file.html
LA_NETL_SONRIS_VI_Dry	Class VI well-points	Sink Nodes	Schooley, C., Pantaleone, S., Shay, J., Strazisar, B., Bauer, J., Morkner, P., Carbon Storage Site Mapping Inquiry Tool, anticipated release 2024, in development. Retrieved from https://arcgis.netl.doe.gov/portal/apps/experiencebuilder/experience/?id=5e0dccd1c2db49e6a8522e9473dfdfb3
LA_SONRIS_Well_density	Class II well points to estimate well density around a 4-mile search radius	Sink Attributes	Louisiana Department of Natural Resources' Strategic Online Natural Resources Information System (SONRIS). (2024). Available at https://sonlite.dnr.state.la.us/ords/f?p=108:2:111681940106496
LA_SONRIS_ClassI_buffer	Class I well points to estimate well density around a 2-mile search radius	Sink Attributes	Louisiana Department of Natural Resources' Strategic Online Natural Resources Information System (SONRIS). (2024). Well Information. https://sonlite.dnr.state.la.us/ords/f?p=108:2:111681940106496:::::
LA_DepEnv_Frio	Depositional environments: Frio	Geological Nodes	Snedden JW, Galloway WE. *The Gulf of Mexico Sedimentary Basin: Depositional Evolution and Petroleum Applications*. Cambridge University Press; 2019.
LA_DepEnv_LowerMio1	Depositional environments: LowerMiocene1	Geological Nodes	Snedden JW, Galloway WE. *The Gulf of Mexico Sedimentary Basin: Depositional Evolution and Petroleum Applications*. Cambridge University Press; 2019.
LA_DepEnv_LowerMio2	Depositional environments:	Geological Nodes	Snedden JW, Galloway WE. *The Gulf of Mexico Sedimentary Basin: Depositional Evolution and Petroleum Applications*. Cambridge University Press; 2019.

	LowerMiocene2		
LA_DepEnv_MiddleMio	Depositional environments: MiddleMiocene	Geological Nodes	Snedden JW, Galloway WE. *The Gulf of Mexico Sedimentary Basin: Depositional Evolution and Petroleum Applications*. Cambridge University Press; 2019.
LA_DepEnv_LowerWil	Depositional environments: LowerWilcox	Geological Nodes	Snedden JW, Galloway WE. *The Gulf of Mexico Sedimentary Basin: Depositional Evolution and Petroleum Applications*. Cambridge University Press; 2019.
LA_DepEnv_MiddleWil	Depositional environments: MiddleWilcox	Geological Nodes	Snedden JW, Galloway WE. *The Gulf of Mexico Sedimentary Basin: Depositional Evolution and Petroleum Applications*. Cambridge University Press; 2019.
LA_DepEnv_UpperWil	Depositional environments: UpperWilcox	Geological Nodes	Snedden JW, Galloway WE. *The Gulf of Mexico Sedimentary Basin: Depositional Evolution and Petroleum Applications*. Cambridge University Press; 2019.
LA_SaltDomes	Salt Domes Shapefiles	Geological Nodes	Wallace, Jr., W.E. 1966. "Fault and Salt Dome Map of South Louisiana." Oversize Map.
LA_SaltDomesBuffer_dist	Salt Domes Buffer Shapefile equivalent to the diameter of the polygon	Geological Nodes	Wallace, Jr., W.E. 1966. "Fault and Salt Dome Map of South Louisiana." Oversize Map.
LA_SurFaults_buffer	Surface Fault Traces with a fixed 500 ft buffer	Geological Nodes	Mclindon, C. 2021. Causes and effects of a late 20th century fault slip event in coastal Louisiana. Available at https://gsa.confex.com/gsa/2021AM/webprogram/Paper369241.html

LA_AccumField_Lower Mio	Oil and Gas Accumulation Fields Boundary: Lower Miocene	Geological Nodes	personal communication – Chris McLindon
LA_AccumField_Middle Mio	Oil and Gas Accumulation Fields Boundary: Middle Miocene	Geological Nodes	personal communication – Chris McLindon
LA_AccumField_Middle Frio	Oil and Gas Accumulation Fields Boundary: Middle Frio	Geological Nodes	personal communication – Chris McLindon
LA_AccumField_UpperFrio	Oil and Gas Accumulation Fields Boundary: Upper Frio	Geological Nodes	personal communication – Chris McLindon
LA_AccumField_Wilcox	Oil and Gas Accumulation Fields Boundary: Wilcox	Geological Nodes	personal communication – Chris McLindon
LA_BRFaults_buffer	Baton Rouge Surface Faults with a 1-mile buffer around them	Geological Nodes	Shen, Z., Dawers, N.H., Törnqvist, T.E., Gasparini, N.M., Hijma, M.P. and Mauz, B. (2017), Mechanisms of late Quaternary fault throw-rate variability along the north central Gulf of Mexico coast: implications for coastal subsidence. Basin Res, 29: 557-570. https://doi.org/10.1111/bre.12184

LA_USDW_Base	Drinking water depth pressure (USDW) Raster	Geological Nodes	Smoot, C. W. (1988). Louisiana hydrologic atlas map no. 3: Altitude of the base of freshwater in Louisiana [Report] (86-4314). (Water-Resources Investigations Report, Issue. U. S. G. Survey. https://pubs.usgs.gov/publication/wri864314
LA_Top_Geopressure	Top Geopressure Raster	Geological Nodes	Burke, L; Kinney, S; Dubiel, R; Pitman, J. (2014). Geopressure Gradient Maps of Southern Louisiana, State and Vicinity. US Geological Survey. Available at https://www.datapages.com/gis-map-publishing-program/gis-open-files/geographic/geopressure-gradient-maps-of-southern-louisiana-state-and-vicinity
LA_FWS_CriticalAreas	U.S. Fish & Wildlife Service Critical Habitat Shapefile	Environmental Nodes	U.S. Fish and Wildlife Service. (2023). FWS HQ ES Critical Habitat [Map]. ArcGIS Online. https://gis-fws.opendata.arcgis.com/maps/794de45b9d774d21aed3bf9b5313ee24
LA_PAD_USGS_ProtectedAreas	United States Geological Service Protected Areas (PAD-US) Shapefile	Environmental Nodes	U.S. Geological Survey. 2022. Protected Areas Database of the United States (PAD-US) 3.0: U.S. Geological Survey data release, https://doi.org/10.5066/P9Q9LQ4B. Retrieved from https://www.usgs.gov/programs/gap-analysis-project/science/protected-areas
LA_NWI_Wetlands	United States National Wetland Inventory Shapefile	Environmental Nodes	U.S. Fish and Wildlife Service. (2024). National Wetlands Inventory [Data set]. U.S. Department of the Interior, Fish and Wildlife Service. https://www.fws.gov/program/national-wetlands-inventory/download-state-wetlands-data
LA_AUS_Nature_Aquifers	Database of Aquifers of the United States	Environmental Nodes	GebreEgziabher, M., Jasechko, S. & Perrone, D. Widespread and increased drilling of wells into fossil aquifers in the USA. Nat Commun 13, 2129 (2022). https://doi.org/10.1038/s41467-022-29678-7
LA_USGS_NHD_Waterbody	USGS National Hydrograph	Environmental Nodes	U.S. Geological Survey. (2023). National Hydrography Dataset. U.S. Department of the Interior. Retrieved from https://prd-

	y Dataset Louisiana Shapefile		tnm.s3.amazonaws.com/index.html?prefix=StagedProducts/Hydrography/NHD/State/GDB/
LA_NETL_SONRIS_Dry	Candidate Dry Hard Wells SONRIS	Candidate Nodes	Louisiana Department of Natural Resources' Strategic Online Natural Resources Information System (SONRIS). (2024). Well Information. https://sonlite.dnr.state.la.us/ords/f?p=108:2:111681940106496:::::

Table A2. Node Attribute Metadata.

Column Name	Group	Variable Description	Process to Join from the GIS Layer	Source GIS Layer	Type of Data
ID	Sink/Candidate	Unique identifier for the reference points	Randomly defined based on Source Data	LA_NETL_SONRIS_VI_Dry	
Latitude	Sink/Candidate	Latitude of the reference point	Reference Points Source Information	LA_NETL_SONRIS_VI_Dry	
Longitude	Sink/Candidate	Longitude of the reference point	Reference Points Source Information	LA_NETL_SONRIS_VI_Dry	
Class	Sink/Candidate	Describe if the reference point is a Class VI well a Class II Shut in Dry Hole or random points.	Reference Points Source Information	LA_NETL_SONRIS_VI_Dry	shapefile points
Refwell_density	Sink Attributes	Count of reference wells within a 4-mile radius	Using the reference layer points as centroids estimate the number of wells (points) from the LA_NETL_SONRIS_VI_Dry layer within a 4-mile radius of each reference point.		
Refwell_DISTm	Sink Attributes	Distance to nearest reference well	Using the reference layer points as centroids estimate the distance from the nearest point from the same reference layer (LA_NETL_SONRIS_VI_Dry) in meters.	LA_NETL_SONRIS_VI_Dry	
Otherwell_density	Sink Attributes	Count all wells within a 4-mile radius	Using the reference layer points as centroids estimate the number of wells (points) from the LA_SONRIS_Well_density layer within a 4-mile radius of each reference point.	LA_SONRIS_Well_density	shapefile points
CrA_Presence	Environmental	Inside/outside critical area	Estimate if a point is spatially inside a polygon on the layer, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_FWS_CriticalAreas	shapefile polygon-lines

CrA_DISTm	Environmental	Distance to the nearest critical area	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_FWS_CriticalArea nearest polygon in meters.		
PAD_Presence	Environmental	Inside/outside protected area	Estimate if a point is spatially inside a polygon on the layer, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_PAD_USGS_ProtectedAreas	shapefile polygon-lines
PAD_DISTm	Environmental	Distance to the nearest protected area	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_PAD_USGS_ProtecedAreas nearest polygon in meters.		
PAD_CAT	Environmental	Category of nearest protected area	Join the column "Category" of the nearest protected area polygon. If there is more than one, create multiple columns with the subindex _1, for example: PAD_CAT_1, PAD_CAT_2		
GW_DISTm	Environmental	Distance to nearest groundwater point	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_NETL_Groundwater_points nearest point in meters	LA_NETL_Groundwater_points	shapefile-points
GW_AquiferDes	Environmental	Aquifer Description of nearest groundwater point	Join the column "NatAqfrDes" of the nearest aquifer polygon.		
GW_DEPTHft	Environmental	Groundwater well depth of the nearest groundwater point in ft	Join the column "WellDepth" of the nearest aquifer polygon		
GW_AquiferName	Environmental	Aquifer Name of the nearest groundwater point	Join the column "LocalAquif" of the nearest aquifer polygon		

WL_Presence	Environmental	Inside/outside wetland area	Estimate if a point is spatially inside a polygon on the layer, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_NWI_Wetlands	shapefile polygon-lines
WL_DISTm	Environmental	Distance to the nearest wetland	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_NWI_Wetlands nearest polygon in meters.		
WL_Type	Environmental	Type of nearest wetland area	Join the column "WETLAND_TY" of the nearest wetland polygon. If there is more than one, create multiple columns with the subindex _1, for example, WL_Type_1, WL_Type_2.		
EJS_TotPop	Socioeconomic	Total Population	Join the column "ACSTOTPOP" of the polygon where the point falls in.	LA_EPA_EJScreen	shapefile polygon
EJS_TractID	Socioeconomic	Census tract ID	Join the column "ID" of the polygon where the point falls in.		
EJS_TractLandm	Socioeconomic	Area of land in the census tract in m2	Join the column "AREALAND" of the polygon where the point falls in.		
EJS_TractWaterm	Socioeconomic	Area of water in the census tract in m2	Join the column "AREAWATER" of the polygon where the point falls in.		
EJS_OZONE	Socioeconomic	Ozone EJ Index	Join the column "D2_OZONE" of the polygon where the point falls in.		
EJS_DIESEL	Socioeconomic	Diesel particulate matter EJ Index	Join the column "D2_DSLPM" of the polygon where the point falls in.		
EJS_ToxicAIR	Socioeconomic	Toxic Releases to Air EJ Index	Join the column "D2_RSEI_AI" of the polygon where the point falls in.		
EJS_TRAFFIC	Socioeconomic	Traffic proximity EJ Index	Join the column "D2_PTRAF" of the polygon where the point falls in.		
EJS_LEADPaint	Socioeconomic	Lead Painting EJ Index	Join the column "D2_LDPNT" of the polygon where the point falls in.		
EJS_Superfund	Socioeconomic	Superfund Proximity EJ Index	Join the column "D2_PNPL" of the polygon where the point falls in.		
EJS_RMP	Socioeconomic	RMP Facility Proximity EJ Index	Join the column "D2_PRMP" of the polygon where the point falls in.		

EJS_HAZWaste	Socioeconomic	Hazardous waste proximity EJ Index	Join the column "D2_PTSDF" of the polygon where the point falls in.		
EJS_UnderTanks	Socioeconomic	Underground storage tanks EJ Index	Join the column "D2_UST" of the polygon where the point falls in.		
EJS_WasteWater	Socioeconomic	Wastewater discharged EJ Index	Join the column "D2_PWDIS" of the polygon where the point falls in.		
EJS_NO2	Socioeconomic	Nitrogen Dioxide (NO2) EJ Index	Join the column "D2_NO2" of the polygon where the point falls in.		
EJS_DrinkWater	Socioeconomic	Drinking Water Non-Compliance EJ Index	Join the column "D2_DWATER" of the polygon where the point falls in.		
NHD_Presence	Environmental	Inside/outside surface water area	Estimate if a point is spatially inside a polygon on the layer, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)		
NHD_DISTm	Environmental	Distance to nearest surface water area	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_USGS_NHD_Waterbody nearest polygon in meters.	LA_USGS_NHD_Waterbody	shapefile polygon-lines
NHD_WBType	Environmental	Type of nearest surface water area	Join the column "FType" of the nearest wetland polygon. If there is more than one, create multiple columns with the subindex _1, for example, NHD_WBType_1, NHD_WBType__2.		
AUS_Presence	Environmental	Inside/outside Aquifer area	Estimate if a point is spatially inside a polygon on the layer, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_AUS_Nature_Aquifers	shapefile polygon
AUS_DISTm	Environmental	Distance to nearest Aquifer	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_AUS_Nature_Aquifers nearest polygon in meters.		

AUS_Name	Environmental	Name of nearest Aquifer area	Join the column "Aquifer" of the nearest aquifer polygon.		
PAR_Presence	Socioeconomic	Inside/outside the parcel	Estimate if a point is spatially inside a polygon on the layer, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_LANDGRID_Parcels	shapefile polygon
PAR_OWNType	Socioeconomic	Owner type	Join the column "own type" of the polygon where the point falls in.		
PAR_SubOWNType	Socioeconomic	Subsurface Owner Type	Join the column "subowntype" of the polygon where the point falls in.		
PAR_Zone	Socioeconomic	Government Zoning Type	Join the column "zoning" of the polygon where the point falls in.		
PAR_ZoneDesc	Socioeconomic	Government Zoning Description	Join the column "zoning_description" of the polygon where the point falls in.		
PAR_STRClass	Socioeconomic	Ownership Structure (Private/public)	Join the column "lbcs_ownership" of the polygon where the point falls in.		
PAR_STRType	Socioeconomic	Type of structure (single-family house, office, etc)	Join the column "lbcs_structure" of the polygon where the point falls in.		
PAR_Areaft	Socioeconomic	Parcel Area in squared feet	Join the column "sqft" of the polygon where the point falls in.		
COAST_Presence	Socioeconomic	Inside/outside coastal zone polygon	Estimate if a point is spatially inside the coastal layer, if it is the value of the column should be inside (value = 1), if it is not inside a polygon the value should be outside (value = 0)	LA_COAST_Zone	shapefile polygon
Dep_Frio_Delta_Presence	Geologic	Inside/outside a Delta System	Estimate if a point is spatially inside the Delta polygons declared in the Delta Column as "Yes", if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_DepEnv_Frio	shapefile polygon
Dep_Frio_Areakm2	Geologic	Area in sqkm in which the System is enclosed	Join the "ENCLOSED_A" value of the polygon where the point falls in		

Dep_Frio_LayerName	Geologic	Layer Name in which the points fall into in the Frio Env	Join the "LAYER" name of the polygon where the point falls in		
Dep_LowerMio1_Delta_Presence	Geologic	Inside/outside a Delta System	Estimate if a point is spatially inside the Delta polygons declared in the Delta Column as "Yes", if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_DepEnv_LowerMio1	shapefile polygon
Dep_LowerMio1_Areakm2	Geologic	Area in sqkm in which the System is enclosed	Join the "ENCLOSED_A" value of the polygon where the point falls in		
Dep_LowerMio1_LayerName	Geologic	Layer Name in which the points fall into in the Lower Miocene 1 Env	Join the "LAYER" name of the polygon where the point falls in		
Dep_LowerMio2_Delta_Presence	Geologic	Inside/outside a Delta System	Estimate if a point is spatially inside the Delta polygons declared in the Delta Column as "Yes", if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_DepEnv_LowerMio2	shapefile polygon
Dep_LowerMio2_Areakm2	Geologic	Area in sqkm in which the System is enclosed	Join the "ENCLOSED_A" value of the polygon where the point falls in		
Dep_LowerMio2_LayerName	Geologic	Layer Name in which the points fall into in the Lower Miocene 2 Env	Join the "LAYER" name of the polygon where the point falls in		
Dep_MiddleMio_Delta_Presence	Geologic	Inside/outside a Delta System	Estimate if a point is spatially inside the Delta polygons declared in the Delta Column as "Yes", if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_DepEnv_MiddleMio	shapefile polygon
Dep_MiddleMio_Areakm2	Geologic	Area in sqkm in which the System is enclosed	Join the "ENCLOSED_A" value of the polygon where the point falls in		
Dep_MiddleMio_LayerName	Geologic	Layer Name in which the points fall into in the Middle Miocene Env	Join the "LAYER" name of the polygon where the point falls in		

Dep_LowerWil_Delta_Presence	Geologic	Inside/outside a Delta System	Estimate if a point is spatially inside the Delta polygons declared in the Delta Column as "Yes", if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_DepEnv_LowerWil	shapefile polygon
Dep_LowerWil_Areakm2	Geologic	Area in sqkm in which the System is enclosed	Join the "ENCLOSED_A" value of the polygon where the point falls in		
Dep_LowerWil_LayerName	Geologic	Layer Name in which the points fall into in the Lower Wilcox Env	Join the "LAYER" name of the polygon where the point falls in		
Dep_MiddleWil_Delta_Presence	Geologic	Inside/outside a Delta System	Estimate if a point is spatially inside the Delta polygons declared in the Delta Column as "Yes", if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_DepEnv_MiddleWil	shapefile polygon
Dep_MiddleWil_Areakm2	Geologic	Area in sqkm in which the System is enclosed	Join the "ENCLOSED_A" value of the polygon where the point falls in		
Dep_MiddleWil_LayerName	Geologic	Layer Name in which the points fall into in the Middle Wilcox Env	Join the "LAYER" name of the polygon where the point falls in		
Dep_UpperWil_Delta_Presence	Geologic	Inside/outside a Delta System	Estimate if a point is spatially inside the Delta polygons declared in the Delta Column as "Yes", if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_DepEnv_UpperWil	shapefile polygon
Dep_UpperWil_Areakm2	Geologic	Area in sqkm in which the System is enclosed	Join the "ENCLOSED_A" value of the polygon where the point falls in		
Dep_UpperWil_LayerName	Geologic	Layer Name in which the points fall into in the Upper Wilcox Env	Join the "LAYER" name of the polygon where the point falls in		
SaltDome_Presence	Geologic	Inside/outside the salt dome polygon	Estimate if a point is spatially inside the Salt Dome polygons, if it is the value of the column should be inside (value=1), if it is	LA_SaltDomes	shapefile polygon

			not inside a polygon the value should be outside (value =0)		
SaltDome_Aream	Geologic	Area m2 of the salt dome	Join the "SHAPE_Area" value of the salt dome polygon where the point falls in		shapefile polygon
BufSaltDome_DIAMm	Geologic	Diameter in meters of the salt dome	Join the "MBG_Diamet" value of the salt dome polygon where the point falls in	LA_SaltDomesBuffer_dist	shapefile polygon
BufSaltDome_Presence	Geologic	Inside/outside the salt dome buffer equivalent to the salt dome diameter	Estimate if a point is spatially inside the Salt Dome polygons, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)		shapefile polygon
SurFaults_Presence	Geologic	Inside/outside the surface faults buffer equivalent to 500 ft from the surface faults.	Estimate if a point is spatially inside the Surface Faults Buffer polygons, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_SurFaults_buffer	shapefile polygon-lines
AccumField_LowMio_Presence	Geologic	Inside/outside the accumulation field polygon	Estimate if a point is spatially inside an accumulation field polygon, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_AccumField_LowerMio	shapefile polygon
AccumField_LowMio_Aream	Geologic	Area m2 of the accumulation field boundary	Join the "SHAPE_Area" value of the accumulation fields polygon where the point falls in		
AccumField_LowMio_DISTm	Geologic	Distance to nearest accumulation field	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_AccumField_LowerMio nearest polygon in meters.		
AccumField_MiddleMio_Presence	Geologic	Inside/outside the accumulation field polygon	Estimate if a point is spatially inside an accumulation field polygon, if it is the value of the column should be inside (value = 1), if it is not inside a polygon the value should be outside (value = 0)	LA_AccumField_MiddleMio	shapefile polygon

AccumField_MiddleMio_Aream	Geologic	Area m2 of the accumulation field boundary	Join the "Shape_Area" value of the accumulation fields polygon where the point falls in		
AccumField_MiddleMio_DISTm	Geologic	Distance to nearest accumulation field	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_AccumField_MiddleMio nearest polygon in meters.		
AccumField_MiddleFrio_Presence	Geologic	Inside/outside the accumulation field polygon	Estimate if a point is spatially inside an accumulation field polygon, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_AccumField_MiddleFrio	shapefile polygon
AccumField_MiddleFrio_Aream	Geologic	Area m2 of the accumulation field boundary	Join the "Shape_Area" value of the accumulation fields polygon where the point falls in		
AccumField_MiddleFrio_DISTm	Geologic	Distance to nearest accumulation field	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_AccumField_MiddleFrio nearest polygon in meters.		
AccumField_UpperFrio_Presence	Geologic	Inside/outside the accumulation field polygon	Estimate if a point is spatially inside an accumulation field polygon, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_AccumField_UpperFrio	shapefile polygon
AccumField_UpperFrio_Aream	Geologic	Area m2 of the accumulation field boundary	Join the "Shape_Area" value of the accumulation fields polygon where the point falls in		
AccumField_UpperFrio_DISTm	Geologic	Distance to nearest accumulation field	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_AccumField_UpperFrio nearest polygon in meters.		

AccumField_Wilcox_Presence	Geologic	Inside/outside the accumulation field polygon	Estimate if a point is spatially inside an accumulation field polygon, if it is the value of the column should be inside (value = 1), if it is not inside a polygon the value should be outside (value = 0)	LA_AccumField_Wilcox	shapefile polygon
AccumField_Wilcox_Aream	Geologic	Area m2 of the accumulation field boundary	Join the "SHAPE_Area" value of the accumulation fields polygon where the point falls in		
AccumField_Wilcox_DISTm	Geologic	Distance to nearest accumulation field	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_AccumField_Wilcox nearest polygon in meters.		
Co2Pipe_DISTm	Geologic	Distance to nearest CO2 pipeline	Using the reference layer points as centroids estimate the nearest distance from the reference points to the theLA_CO2Pipe lines in meters	LA_CO2Pipe	shapefile polygon-lines
CO2Source_DISTm	Geologic	Distance to nearest CO2 source in meters	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_CO2_Source nearest point in meters	LA_CO2_Source	shapefile points
CO2Source_nearTonsCO2e	Geologic	Nearest Source GHG Quantity of emissions (CO2e)	Join the "GHG QUANTITY (METRIC TONS CO2e)" value of the nearest LA_CO2_Source point.		
CO2Source_density	Geologic	Count of sources within a 4-mile radius (Source density)	Using the reference layer points as centroids estimate the number of facilities (points) from the LA_CO2_Source layer within a 4-mile radius of each reference point.		
CO2Source_SumDensTons	Geologic	SUM of GHG emissions (CO2e) within a 4-mile radius	Sum the "GHG QUANTITY (METRIC TONS CO2e)" column value of all the points in the LA_CO2_Source layer within a 4-mile radius of each reference point.		
Tribal_DISTm	Socioeconomic	Distance to nearest tribal land in meters	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_Tribal_land nearest polygon in meters	LA_Tribal_land	shapefile polygon

Pet_ResDISTm	Geologic	Distance to nearest strategic petroleum reserve in meters	Using the reference layer points as centroids estimate the nearest distance from the reference points to the LA_Pet_Reserve nearest line in meters	LA_Pet_Reserve	shapefile points
BRFaults_Presence	Geologic	Inside/outside the Baton Rouge surface faults buffer equivalent to 1 mile (1.6 km) from the BR surface faults.	Estimate if a point is spatially inside the LA_BRFaults_buffer polygons, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_BRFaults_buffer	shapefile polygon-lines
ClassIwell_Presence	Geologic	Inside/outside the LA_SONRIS_ClassI_buffer is equivalent to 2 miles.	Estimate if a point is spatially inside the LA_SONRIS_ClassI_buffer polygons, if it is the value of the column should be inside (value=1), if it is not inside a polygon the value should be outside (value =0)	LA_SONRIS_ClassI_buffer	shapefile points
USDW_valuem	Geologic	Value of the Drinking water base in meters	Join the raster value of the LA_USDW_Base pixel where the point falls in	LA_USDW_Base	raster
USDW_meanm	Geologic	The mean value of the Drinking water base (meters) within a 4-mile radius	Estimate the mean LA_USDW_Base pixel value within a 4-mile radius of each reference point.		
TopGeo_value	Geologic	Value of the top geopressure	Join the raster value of the LA_Top_Geopressure pixel where the point falls in	LA_Top_Geopressure	raster
TopGeo_mean	Geologic	The mean value of the top geopressure within a 4-mile radius	Estimate the mean LA_Top_Geopressure pixel value within a 4-mile radius of each reference point.		
Geo_complex_score	Geologic	Score of geological complexity of a location	Join the "score" value of the geological complexity polygon where the point falls in	LA_Geological_complex	shapefile polygon

Table A3. Edgelist and Edge Attributes Metadata

Column Name	Relationship 1 (Sink-to-Sink)	Relationship 2 (Sink-to-Candidate)	Relationship 3 (Sink/Candidate-to-Population)	Relationship 4 (Sink/Candidate-to-Source)	Relationship 5 (Sink/Candidate-to-Pipeline)
From column	Sink Node ID	Sink Node ID	Sink Node ID + Candidate Node ID	Sink Node ID + Candidate Node ID	Sink Node ID + Candidate Node ID
To column	Sink Nodes ID	Candidate Node ID	Population Node ID	Source Node ID	Pipeline Node ID
Relational process description	Calculate all-to-all distances between sink nodes. Paste sink ID in "from" and "to" columns, estimate distance in km for all point-to-point combinations. Output: Dist_Km	Calculate one-to-many distances from the sink to candidate nodes. Paste sink ID in "from" and candidate ID in "to" columns, estimate distances in km (no candidate-to-candidate distances needed). Output: Dist_Km	Calculate distances from sink or candidate nodes to population centroids within a 30-mile radius. Paste the sink or candidate ID in the "from" column and the FIPS in the "to" column. Output: Dist_Km, Tract_Densqmi (from POP_SQMI), and TotPop (from POPULATION).	Calculate distances from sink/candidate nodes to source centroids within a 100-mile radius. Paste sink/candidate ID in "from" and GHGRP_ID in "to" columns. Output: Dist_Km and TonsCO2 (from GHG_QUAN_5).	Calculate all-to-all distances from sink/candidate nodes to nearest pipeline sections. Paste sink/candidate ID in "from" and Pipeline ID in "to" columns. Output: Dist_Km, Pipe_Diam (from DIAMETER), and Pipe_Length (from LENGTH_MIL).
Edge Attribute Column Names	Dist_Km	Dist_Km	Dist_Km, Tract_Densqmi, TotPop	Dist_Km, TonsCO2	Dist_Km, Pipe_Diam, Pipe_Length
Type	Sink	Candidate	PopulationCluster	Source	Pipeline